

ORANGE COUNTY AP, TX, USA

WMO: 723629

Lat: 30.069N

Lon: 93.804W

Elev: 4

StdP: 101.28

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	0.3	-7.4	2.0	5.0	-4.8	2.5	7.7	8.4	14.1	7.7	13.8	1.4	340	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	34.4	25.6	33.3	25.4	32.6	25.3	27.9	31.1	27.3	30.4	26.8	29.8	2.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.0	29.2	26.6	22.1	28.6	26.1	21.5	28.3	88.7	31.1	86.0	30.1	83.8	29.9	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	7.0	5.9	-3.3	36.2	1.9	1.3	-4.7	37.1	-5.8	37.9	-6.8	38.6	-8.2	39.5		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-4.2	28.0	2.1	0.9	-5.7	28.7	-7.0	29.2	-8.1	29.7	-9.6	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	10.4	13.4	17.0	20.0	24.0	27.1	27.8	28.2	25.8	21.0	15.9	12.3
	DBStd	7.20	4.88	5.29	4.56	3.52	2.82	1.69	1.47	1.46	2.33	4.16	4.99	5.11
	HDD10.0	127	54	25	6	1	0	0	0	0	0	0	8	33
	HDD18.3	827	247	150	81	24	3	0	0	0	0	22	104	197
	CDD10.0	3884	68	121	222	302	434	513	551	565	475	342	186	106
	CDD18.3	1542	3	12	38	75	178	263	293	307	225	105	33	11
	CDH23.3	14460	3	22	108	390	1453	2821	3188	3392	2029	881	145	28
	CDH26.7	5447	0	1	4	32	371	1149	1359	1504	783	232	10	0
Wind	WSAvg	2.4	2.7	3.0	3.0	3.0	2.7	2.2	1.7	1.7	1.7	2.1	2.3	2.6
Precipitation	PrecAvg	1499	138	99	96	105	120	142	140	120	139	143	120	131
	PrecMax	2014	341	206	204	199	297	315	308	254	381	498	276	310
	PrecMin	975	18	20	19	15	3	8	41	15	29	3	20	64
	PrecStd	281	78	58	45	57	78	79	71	71	85	135	69	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	25.6	27.2	28.8	32.4	35.6	34.8	36.1	35.1	32.3	27.9	25.7
		MCWB	18.5	20.8	19.8	20.1	22.4	24.7	24.7	25.0	25.9	24.9	23.0	21.3
	2%	DB	22.0	23.9	25.8	27.6	30.9	33.0	33.8	34.5	33.2	30.6	26.3	24.0
		MCWB	17.8	20.0	20.3	21.0	23.4	24.6	25.1	26.2	25.2	23.7	21.5	20.9
	5%	DB	20.2	22.5	24.8	26.7	29.9	32.4	32.9	33.4	32.2	29.1	25.1	22.5
		MCWB	17.3	19.6	19.9	20.9	23.5	24.7	25.2	26.0	24.8	23.2	20.6	20.0
	10%	DB	18.7	21.1	23.1	25.8	28.8	31.6	32.2	32.6	31.0	27.7	23.6	21.0
		MCWB	16.2	18.9	19.2	20.6	23.3	24.8	25.2	25.9	24.5	22.0	20.4	19.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	22.8	22.8	24.6	26.4	28.0	28.0	29.0	27.8	26.9	24.2	22.9
		MCDB	22.0	24.1	25.2	26.4	29.1	30.5	31.1	32.6	31.1	29.9	26.5	24.5
	2%	WB	19.7	21.8	22.0	23.5	25.6	27.0	27.4	28.2	27.0	26.0	23.1	21.9
		MCDB	20.6	23.0	24.0	25.6	28.3	29.7	30.3	31.6	30.2	28.5	25.1	23.1
	5%	WB	18.4	20.8	21.4	22.8	25.0	26.5	26.8	27.5	26.5	25.2	22.4	21.0
		MCDB	19.3	21.7	23.1	25.0	27.7	29.5	29.8	30.8	29.5	27.4	24.2	22.1
	10%	WB	16.9	19.6	20.8	22.2	24.3	25.9	26.4	27.0	25.8	24.2	21.3	19.3
		MCDB	18.0	20.5	22.3	24.3	27.2	29.2	29.3	30.0	28.6	26.2	22.9	20.5
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.0	10.0	9.9	8.9	8.6	8.6	8.7	9.5	11.3	11.3	10.2
		MCDBR	11.5	9.6	9.0	9.7	9.4	9.9	9.6	9.8	10.2	10.5	10.3	10.6
		MCWBR	8.4	6.3	5.0	4.7	3.7	3.3	3.4	3.3	3.8	4.7	6.1	7.6
	5% WB	MCDBR	10.1	8.0	7.5	7.4	7.3	8.0	8.7	8.4	8.6	9.2	8.5	9.6
		MCWBR	8.8	6.5	5.1	4.7	3.4	3.4	3.7	3.4	3.6	5.6	6.1	7.9
Clear-Sky Solar Irradiance	taub	0.340	0.357	0.375	0.430	0.471	0.499	0.500	0.483	0.456	0.384	0.371	0.351	
	taud	2.524	2.484	2.425	2.274	2.190	2.150	2.189	2.239	2.279	2.467	2.444	2.495	
	Ebn at Noon	894	907	910	867	829	801	799	810	820	867	848	861	
	Edh at Noon	88	100	113	136	148	154	147	139	129	100	95	87	
All-Sky Solar Radiation	RadAvg	2.86	3.24	4.51	5.44	6.09	6.20	5.96	5.65	4.99	4.37	3.30	2.53	
	RadStd	0.28	0.54	0.40	0.39	0.41	0.47	0.48	0.49	0.50	0.56	0.32	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+85

Nomenclature: See separate page